SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

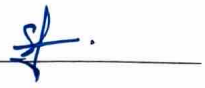
Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**

Question Count: 4

Code of Conduct

I S.SHAMAH ALI FATHIMA (192573002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Cloud Computing Fundamentals.

Definition:

Cloud Computing is the delivery of computing resources such as servers, storage, software, database, networking over the Internet.

Uses:

* Online file storage
* Hosting website, application
* Data backup
* Online education.

Characteristics.

* One demand service
* Scalability
* Resource sharing.

Advantages.

* Reduces hardware cost
* Easy to access from anywhere
* Automatic updates

Explanation:

* It allows the users to access resources whenever they need them.
* It is scalable, flexible and usually follows a pay as you use mode.
* It is used for data storage, backup, application hosting and online services.

2)

Evolution of cloud Computing.

Definition

Evolution of cloud computing refers to the development of computing technology from traditional hardware system to Internet-based cloud services.

Internet Evolution

* Internet
* Web Application
* Web 2.0
* Online Software

Hard Ware Evolution

* Mainframes
* Personal Computers
* Servers
* High-performance Computers.

Advantages

* Higher computing power
* Better resource sharing
* Lower infrastructure cost.

Explanation:

* The growth of the Internet made it possible to access software and resources remotely.

* Web technologies further improved online application & services.

* These developments led to service-based models such as IaaS, PaaS, SaaS.

Cloud service Model.

Definition

Cloud service Models describe the different types of computing services provided to users through the cloud.

Uses

* IaaS : Provides virtual servers
* PaaS : testing application
* Xaas : Provides different resources

Advantages

* Reduces hardware and maintenance costs.
* Easy to access from anyplace
* Supports scalability & flexibility.

Main Models.

* IaaS → Infrastructure
* PaaS → Platform
* SaaS → Software
* XaaS → Everything as service.

Virtualization of Web Services as Enablers of Cloud

Definition:

Virtualization creates virtual versions of physical computing resources while Web services all different applications.

Uses:

* Server consolidation
* cloud infrastructure
* Testing & Development

Advantages:

* Better hardware utilization
* Reduces hardware cost
* Easy resource allocation.

Enabling cloud.

* Virtualization → Cloud Deployment
* Web services → Cloud communication.

Explanation:

3) Cloud Service Models:

* Iaas → Provides infrastructure such as virtual machine, storage & networks.
* Paas → Provides a platform for deploying, testing & developing applications.
* Saas → Provides ready to use software through the Internet.
* Xaas → Provides various IT resources such as storage.

4) Virtualization of web services:

* Virtualization creates virtual version of servers, storage & other computing resources.
* It allows multiple virtual machines to run on a single physical server.
* This improves resources utilization & reduces infrastructure costs.
* Web services allow different applications to communicate and exchage data over the Internet.